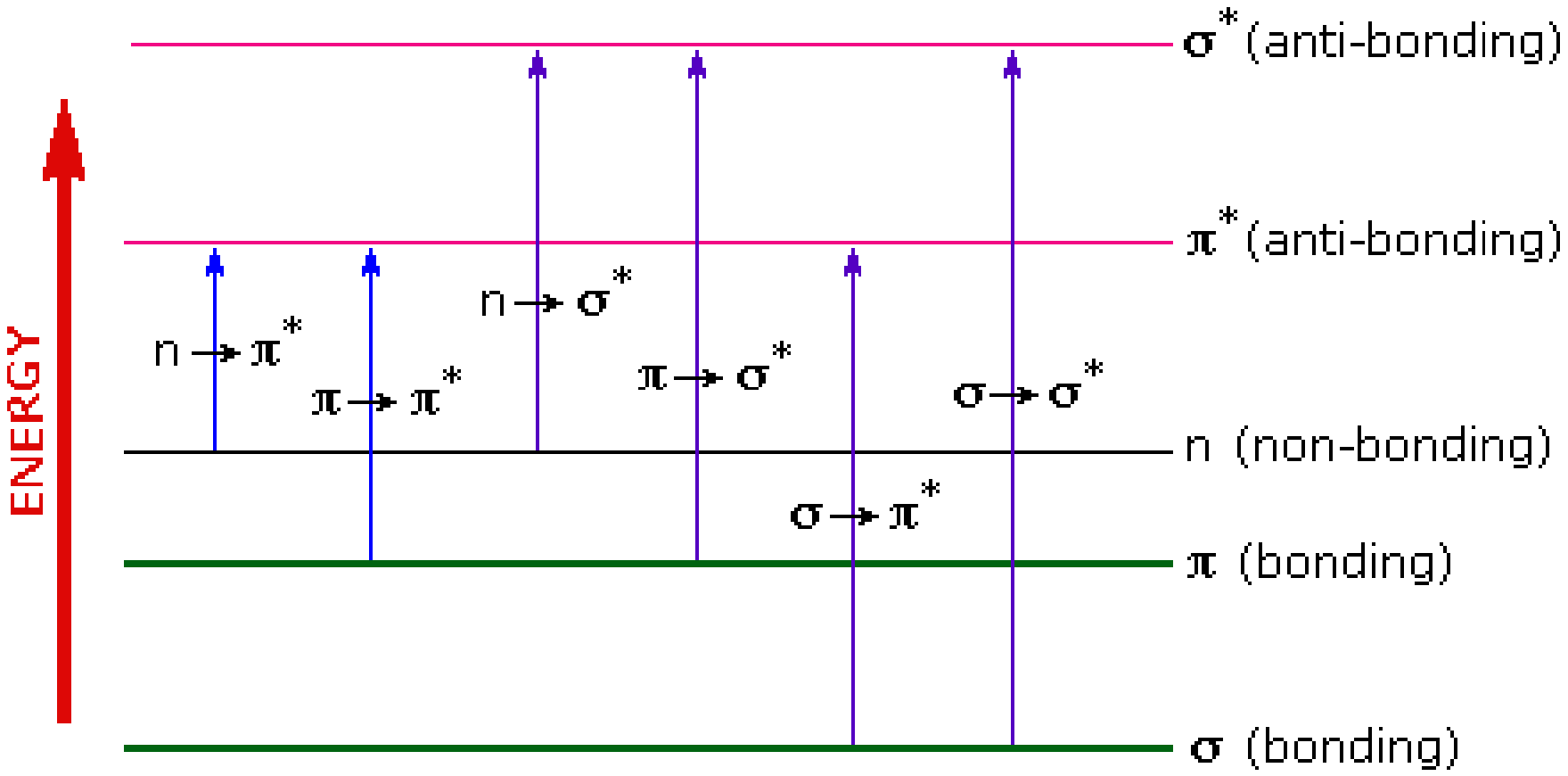
Electronic Spectroscopy



Selection Rule

The transition between the possible electronic energy states is governed by certain selection rules.

In an electronic transition:

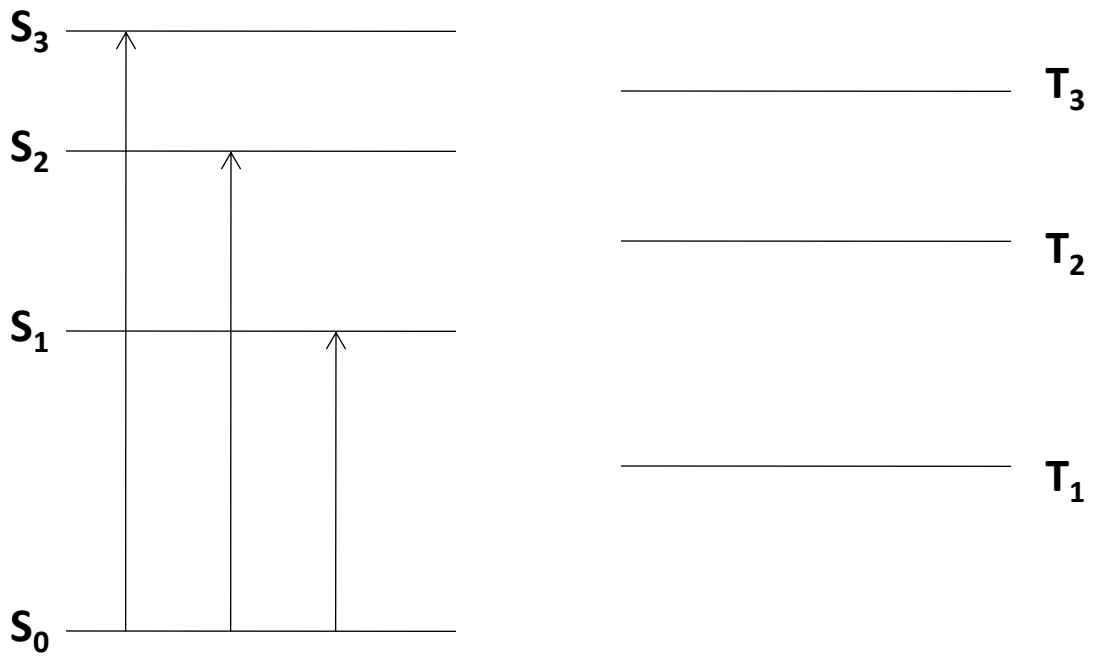
(i) there is no restriction on changes in n; Δn = any value
(ii) S can combine with its own value; $\Delta S = 0$
(iii) L can vary by 0 or ± 1 unit; $\Delta L = 0, \pm 1$
(iv) J can vary by 0 or ± 1 except that J=0 to J=0 transition is not allowed; $\Delta J = 0, \pm 1$ (except $0 \rightarrow 0$)

Laporte's Rule

Notation for Excited States of Organic Molecules

The most common way to denote the excited states is by their multiplicities, S (singlet) or T (triplet).

The ground state is denoted as S_0 and the higher excited states as S_1, S_2, S_3, T_1, T_2, etc.



Michael Kasha has evolved another system of notation in which **electronic states are expressed in terms of the initial and final orbitals involved in the transition**.

In general, four types of molecules are identified:

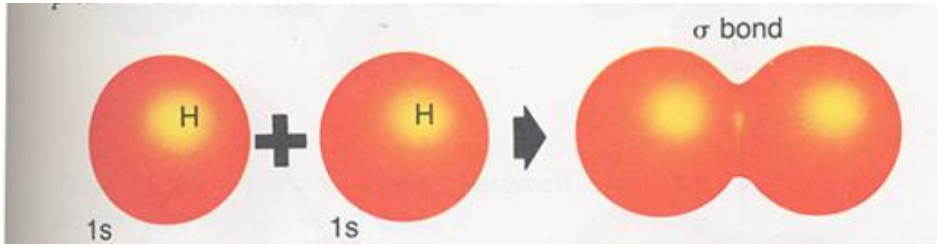
(i) Saturated molecules with σ-molecular orbitals (MOs) only.
Ex – Paraffin hydrocarbons, B_2H_6.

(ii) Saturated molecules with σ and non-bonding π-MOs.
Ex – H_2O, NH_3, CH_3I.

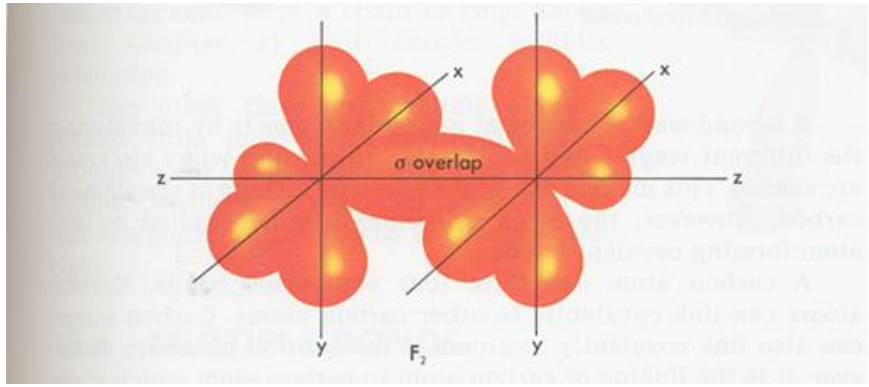
(iii) Unsaturated molecules with σ and π-MOs.
Ex – $H_2C{=}CH_2$, $HC{\equiv}CH$

(iv) Unsaturated molecules with σ, π and n-MOs.
Ex – aldehydes, ketones, pyridines, amines and other heterocyclic organic compounds containing O, N, S, etc.

The σ-MOs are bonding type with axial symmetry.



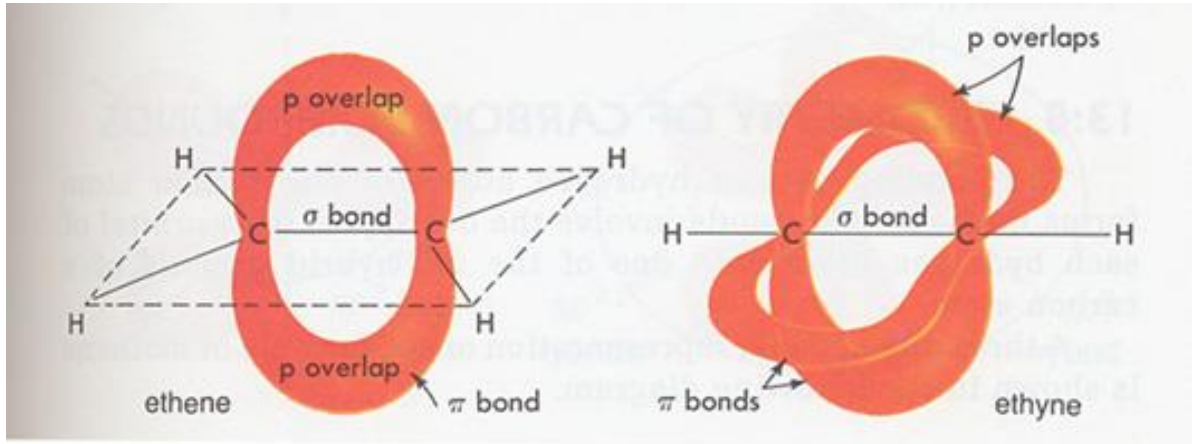
They are formed by overlap of s or p_z orbitals or hybrid orbitals like *sp*, sp^2, sp^3, etc. along the direction of the bond.



They form strong single bonds with two spin paired electrons localized between the combining atoms.

The π-MOs have a nodal plane in the plane of the molecule.

They form by the overlap of p_x, p_y or hybridized orbitals of similar symmetry.



They participate in double or triple bonds using a pair of electrons with antiparallel spin.

The π-bonds are less strong than σ-bonds but give rigidity to the molecule.

σ^* and π^*-MOs are **antibonding** equivalents of σ and π-MOs, respectively.

The *n* MOs are in general, pure atomic orbitals and do not take part in construction of MO $\rightarrow$ **nonbonding**.

The pair of electrons occupying these orbitals is called **lone pair electrons or nonbonding electrons**.

The *lone pair electrons* have far-reaching importance in H-bond formation which is responsible for the native structure of biomolecules such as DNA.

Classification of Electronic States

Similar to the electronic states of atoms, a number of electronic states is also present in the molecules and this gives rise to many electronic bands.

Similar to atomic states, defined by four quantum numbers (n, l, m_l, s), molecular states are also dependent on the four quantum numbers.

Due to poor applicability of the Schrödinger wave equation to the molecules, some of the quantum numbers are not well-defined as those of atoms.

Orbital Angular Momentum

- The motion of the electrons **in an atom** is described in presence of spherically symmetrical field.

- As a result of this motion, the electronic orbital angular momentum, ***L***, possesses a constant value as long as it interacts weakly with the spin angular momentum.

- On the other hand, in case of diatomic molecules, the force field is no longer spherically symmetrical, but possesses a cylindrical symmetry about the internuclear or molecular axis → **it has axis of symmetry**

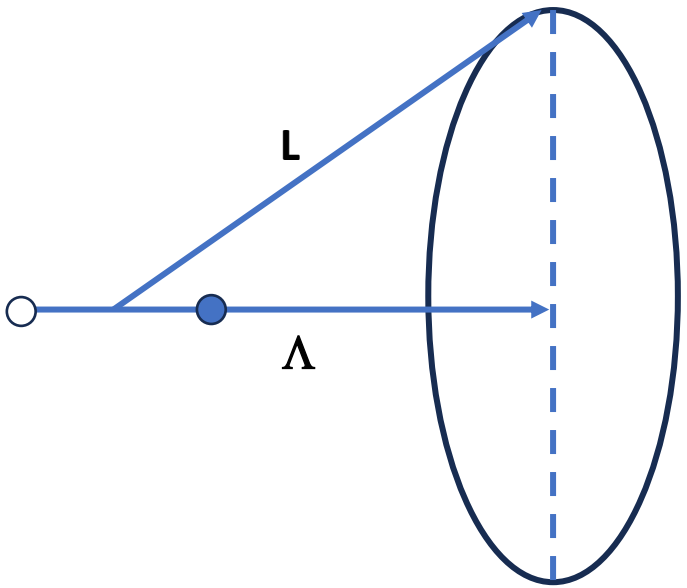
- As a result of this, only the component of the orbital angular momentum of the electrons along the molecular axis will be defined and is a **constant of motion**.

The precession of **L** around the field direction (molecular axis) with a component of constant value of M_L.

The values of M_L can be: M_L = L, L-1, L-2 ……., -L

As the electric field strength increases, the orbital angular momentum **L** precesses faster about the molecular axis.

The component of the angular momentum around the molecular axis (**M_L**) remains well-defined.



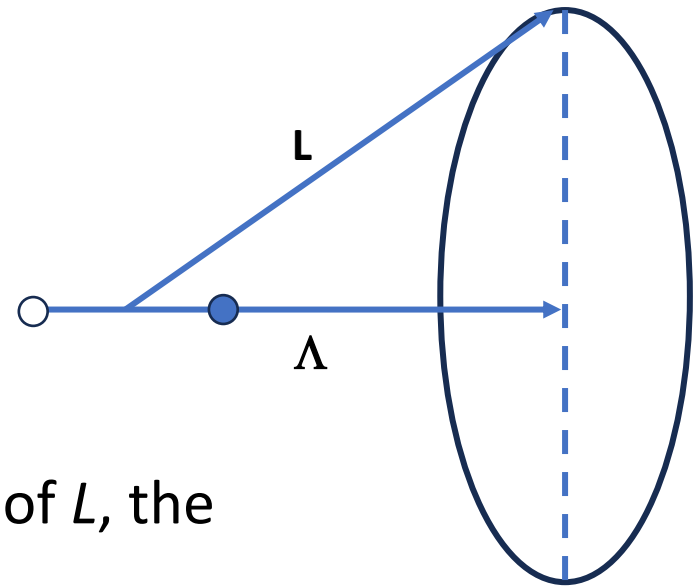
The electronic states of diatomic molecules are defined by the value of the component of the electronic angular momentum (M_L).

The **component of electronic orbital angular momentum around the molecular axis** is defined as

$$\Lambda = |M_L|$$

and the values of Λ for a given value of L can be

$$\Lambda = 0,1,2,3, \dots \dots \dots L$$



Similar to the nomenclature of atomic states by the value of *L*, the molecular states based on the values of Λ are defined as

Λ	0	1	2	3….
States	Σ	Π	Δ	Φ….

The electronic states, except for Σ, are **doubly degenerate**, because each M_L can have two values along the molecular axis, whereas all the Σ states are **non-degenerate**.

Spin Angular Momentum

The spin of the individual electron can be added to get the total spin value ***S***.

Its value can be integral or half-integral, depending on whether the number of electrons is even or odd.

The notation used for the resultant value of spin is the same as that for atoms, i.e., *S* and thus the resultant spin angular momentum is

$$S = \sqrt{S(S+1)}\frac{h}{2\pi}$$

where the total spin quantum number is given as

$$S = \sum_i S_i, \sum_i S_i - 1, \sum_i S_i - 2, \ldots \ldots .. 2,1,0$$

The component of spin angular momentum along the molecular axis is defined by the interaction between the spin motion and the magnetic field produced by the motion of electrons in the molecule around the molecular axis.

- In case of Σ state (where the electric field is zero), the resultant S is unaffected in space, unless the molecule is rotating and there is no magnetic field.

- When the value of $\Lambda \neq 0$ (i.e., Π, Δ, Φ, etc.), the orbital motion of the electrons around the molecular axis will create internal magnetic field in the direction of molecular axis.

- This magnetic field causes the precession of the spin S along the field direction and will produce a constant component, M_S, denoted by σ for a single electron or Σ for more than one electron.

The values of Σ, can be: $\Sigma = S, S-1, S-2, \dots \dots \dots, -S$. This means that there will be 2S+1 different possible values of Σ. Σ can have positive as well as negative values.

Total Angular Momentum

In case of molecules, an algebraic addition of Λ and Σ since both the vectors lie along the molecular axis joining the nuclei.

The total angular momentum around the molecular axis, denoted by Ω, can be obtained by the addition of Λ and Σ.

$$\Omega = |\Lambda + \Sigma|$$

Λ and Σ may have the same or opposite direction along the molecular axis.

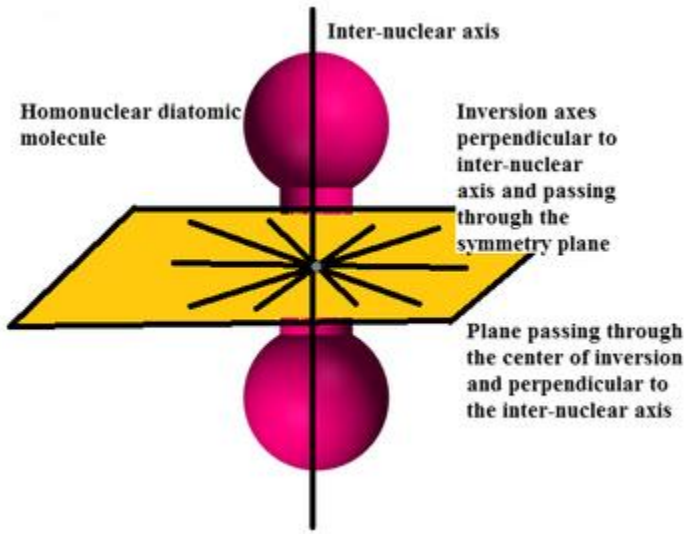
- If $\Lambda \neq 0$, there will be 2S+1 different values of $\Lambda + \Sigma$ for a given value of Λ, i.e. $\Lambda \geq \mathrm{S}$.
- 2S+1 is called multiplicity of a state and it is independent of the value of Λ.

The term symbol of a molecule can be written as: ${}^{2S+1}|\Lambda|_{\Omega}$

Symmetry properties of the electronic wave functions or orbitals

Two Symmetry Operations:

(a) Centre of inversion
(b) Reflection through symmetry plane



The "Centre of inversion" symmetry operation gives rise the subscript *g* or *u*, which indicates whether the orbital changes sign or not when an inversion operation passes through the centre of symmetry. This is called "parity".

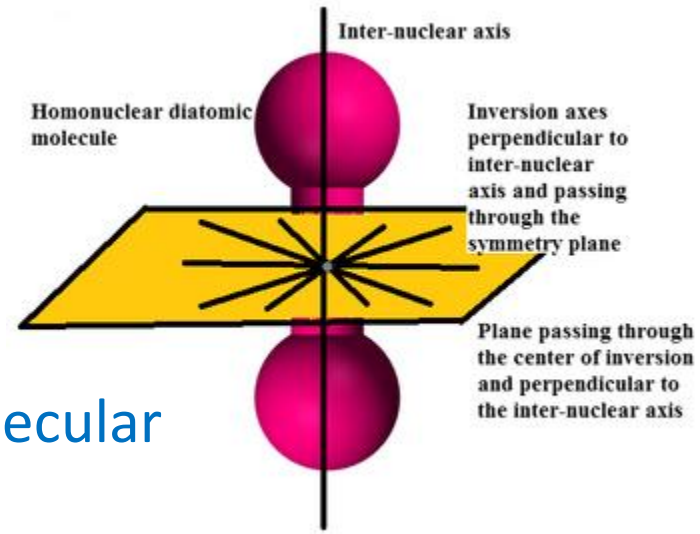
The overall parity, in a homonuclear diatomic molecule, is obtained by multiplying the parities of individual electron.

gerade x gerade = gerade (g x g = g)
ungerade x ungerade = gerade (u x u = g)
ungerade x gerade = ungerade (u x g = u)

Symmetry properties of the electronic wave functions or orbitals

Two Symmetry Operations:

(a) Centre of inversion
(b) Reflection through symmetry plane



In a diatomic molecule, any plane passing through the molecular axis is a plane of symmetry.

A non-degenerate Σ state <u>may or may not change sign</u> when reflected through a plane passing through both the nuclei.

(i) In the former case the state is called Σ^-
(ii) In the latter case it is called Σ^+

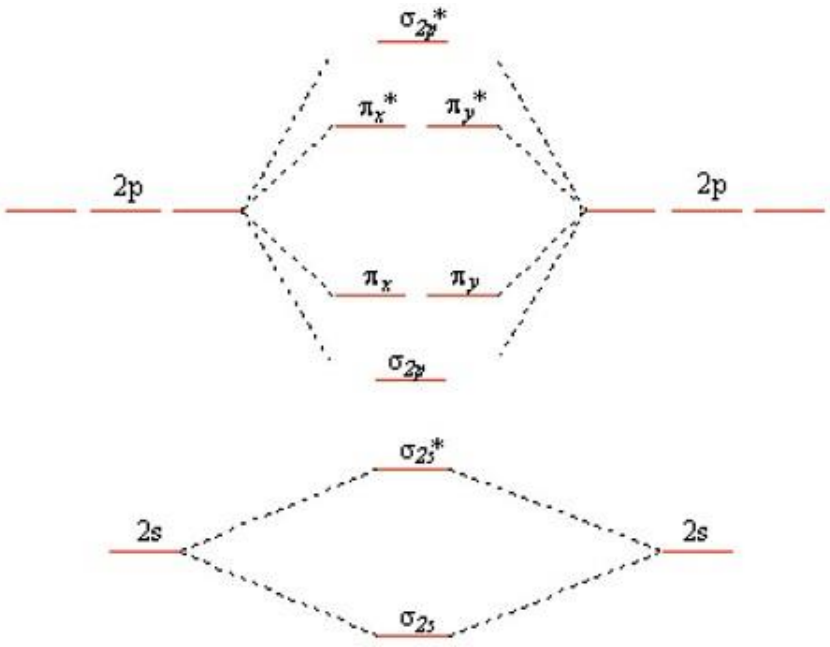
Term Symbols

How to calculate the Term Symbols?

1) Write the electronic configuration using the energy levels.

2) All the completely filled molecular orbitals (e.g., σ^2, π^4, etc.) give rise to $^1\Sigma_g^+$ $\rightarrow$ because the resultant orbital angular momentum of a closed shell is always zero and the total spin is also zero $\rightarrow$ the **overall parity** will be *g*.

3) The term symbol of electronic configuration like π^1 or π^3 i.e., **unfilled orbitals**, are the same.



Calculate the term symbol for the electronic configuration σ^1.

Here, the electron is present in the σ orbital.

$\lambda = \Lambda = 0$, $s = S = 1/2$ and $2S+1 = (2x1/2)+1=2$

Thus, the state is a doublet and the parity is *g*.

$\Omega = \Lambda + \Sigma = 0 + 1/2 = ½$

The term symbol is, $^2\Sigma_{1/2(g)}$

Consider O_2 molecule with electronic configuration:
$KK\sigma_{2s}^2\sigma_{2s}^{*2}\sigma_{2pz}^2\pi_{2px}^2\pi_{2py}^2\pi_{2px}^{*1}\pi_{2py}^{*1}$. Calculate the term symbols.

Since, **$KK\sigma_{2s}^2\sigma_{2s}^{*2}\sigma_{2pz}^2\sigma_{2px}^2\sigma_{2py}^2$** are completely filled MOs $\rightarrow$ contributions to the total angular momentum component along the molecular axis will be zero.

- O_2 molecule has two unpaired electrons.

Here, λ_1 = +1 or -1, λ_2 = +1 or -1

Λ = +2,-2,0

λ_1	+1	-1	λ_1	+1	-1
λ_2	+1	-1	λ_2	-1	+1
Λ	+2	-2	Λ	0	0

Two values of Λ obtained (+2 or -2) are consistent with the fact that $\Lambda \neq 0$, are **doubly degenerate**.

Λ = 0 also correspond to two different ways of arranging λ_i.

In other words, Λ = 0 is also doubly degenerate, <u>which is not possible</u>.

Remember:

A non-degenerate Σ state <u>may or may not change sign</u> when reflected through a plane passing through both the nuclei.

(i) In the former case the state is called Σ^-

(ii) In the latter case it is called Σ^+

- For Λ = +2 or -2, both the electrons are present in the same MO.

Thus, s_1 = +1/2, and s_2 = -1/2, leading to S = 1/2-1/2 = 0, 2S+1 = 2x0+1 = 1

$$\Omega = 2+0, \text{ or } 2-0 = 2$$

The term symbol is $^1\Delta_2$.

- For Λ = 0, s_1 = +1/2, s_2 = +1/2 and S = 1, 2S+1 = 3

  s_1 = +1/2, s_2 = -1/2 and S = 0, 2S+1 = 1

$$\Omega = 0+1 = 1$$
$$\Omega = 0+0 = 0$$

Term symbols can be $^3\Sigma^-$ and $^1\Sigma^+$.

Since, the **total wave function** ($\Psi_T = \Psi_{space}\Psi_{spin}$) should be <u>anti-symmetric</u>, so anti-parallel spins of electrons has to be combined with Σ^+ (i.e., $^1\Sigma^+$) and parallel spins with Σ^- (i.e., $^3\Sigma^-$).

Further, both the electrons in π^* orbitals have *g*-parity, the total parity will be *g*x*g* = *g*.

The energy level diagrams or spectroscopic states can be based on Hund's rule, i.e., the ground state should have the highest multiplicity.
Further, when two electrons with parallel spin are present in π or δ MOs, it gives rise to Σ^- state.
This means that $^3\Sigma_g^-$ is the ground state.

The complete energy level diagram is:

$^1\Sigma g^+$ _____

$^1\Delta g$ _____

$^3\Sigma g^-$ _____